

Q4a in the Proper Context: The Holographic Pair is Not Selected

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Abstract

Paper 11 left ultraviolet completion open by construction and refused to invent a CFT. That refusal was correct as a ban on fabrication. It is not the end of the question the program actually asked.

The monograph states the question precisely: the New Standard Model is emergent book-keeping; the microscopic description is the boundary theory of a holographic pair; no first-principles selection of that pair exists [?]. That is Q4a. It is a *selection* problem, not an invitation to name $\mathcal{N} = 4$ super-Yang–Mills or a Calabi–Yau.

This note answers the selection problem. The certified principles (Casini–Huerta–Myers, Faulkner–Guica–Hartman–Myers–Van Raamsdonk, the Paper II modular rule) determine the bulk *gravitational* sector given a pair. They are blind to G_{SM} , the number of generations, and the Yukawas. They therefore underdetermine the pair. Independently, the Standard Model cannot *be* the boundary CFT: its one-loop coefficients are $(b_1, b_2, b_3) = (41/10, -19/6, -7)$, all nonzero, and it has 118 light on-shell degrees of freedom, not a large- N gap [?].

Existence of some other pair whose bulk infrared is the NSM remains [O]. No pair is constructed. Final Status “UV completion” is not moved to [P].

Standard of proof

Q4a, in the vocabulary of this program, is the following pair of questions.

1. **Selection.** Do the certified entanglement principles single out a holographic pair (CFT, bulk) of which (??) is the bookkeeping?
2. **Existence.** Does at least one such pair exist?

A named CFT offered without a derivation from those principles is a fabrication [?]. A uniqueness theorem, or a proof that uniqueness fails, is an answer to (1). Item (2) is not solved by solving (1).

1 What the pair is

The NSM action is the bulk effective theory

$$S_{\text{NSM}} = \frac{1}{4\kappa} \int \varepsilon_{abcd} e^a \wedge e^b \wedge R^{cd} - \frac{\Lambda}{6} \int \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d + \int e \mathcal{L}_{\text{SM}}. \quad (1)$$

In the certified regime the microscopic description is a holographic pair [?, ?]: a unitary CFT on the conformal boundary of AdS, with a bulk dual whose infrared is (??). Two readings must not be mixed.

Definition 1.1 (Reading A, the monograph). The Standard Model fields in (??) are *bulk* fields. The boundary CFT is some other theory, with a global flavour symmetry whose bulk gauge fields are G_{SM} .

Definition 1.2 (Reading B, the temptation). The Standard Model *is* the boundary CFT, and bulk Einstein–Cartan is its entanglement geometry.

Reading B would be a selection. It is false, by the next section. Reading A is the monograph’s reading. Under it, Q4a is: which CFT?

2 The Standard Model is not a holographic CFT

A necessary condition for a four-dimensional gauge theory to be a CFT is the vanishing of all one-loop beta-function coefficients in the convention $\beta_i = b_i g_i^3 / (16\pi^2)$. From the SM representation table, with $Q = T_3 + Y/2$ and GUT-normalized $U(1)$,

$$b_1 = \frac{41}{10}, \quad b_2 = -\frac{19}{6}, \quad b_3 = -7. \quad (2)$$

All three are nonzero. The Standard Model is not a CFT. [P] (*computed* from Dynkin indices; textbook values recovered).

A necessary condition for a *classical* Einstein dual is a parametrically large N and a sparse spectrum of light operators with a large gap [?]. Before electroweak symmetry breaking the SM has 118 on-shell light degrees of freedom (24 gauge helicities, 4 Higgs, 90 fermionic). That is $O(10^2)$, not a limit $N \rightarrow \infty$. [P] (count).

Theorem 2.1 (Reading B is false). *The Standard Model is not the boundary theory of a holographic pair with a classical Einstein–Cartan bulk. [P] (necessary conditions fail).*

A mutation confirms the CFT check is not tautological: $\mathcal{N} = 4$ super-Yang–Mills has one-loop $b = 0$ by the adjoint index identity $T(\text{adj}) = C_2(G)$. The same script that rejects the SM accepts that necessary condition for $\mathcal{N} = 4$. Accepting the necessary condition is not a selection of $\mathcal{N} = 4$ as the pair.

3 What the certified principles actually see

The Casini–Huerta–Myers modular Hamiltonian of a ball of radius R in a CFT vacuum is [?]

$$H_B = 2\pi \int_{|x| < R} d^{d-1}x \frac{R^2 - |x|^2}{2R} T_{00}(x). \quad (3)$$

Faulkner–Guica–Hartman–Myers–Van Raamsdonk convert the first law $\delta S_B = \delta \langle H_B \rangle$ into linearized Einstein equations sourced by $\delta \langle T_{\mu\nu} \rangle$ [?]. The only CFT data in that argument are the stress tensor, the central charge that normalizes it, and the ball.

Paper II selects the Young symmetry of an explicit spin-current source: mixed-symmetry non-minimal couplings are relevant deformations that destroy the conformal frame; the surviving sector is axial and minimal [?]. That is a statement about *how* a spinor couples to ω , not about *which* gauge group the spinor is charged under.

Theorem 3.1 (Underdetermination). *Let $\mathcal{P}$ be the set of holographic pairs whose bulk infrared contains Einstein gravity, a cosmological term, and some minimally coupled anomaly-free chiral gauge theory with an axial current. The maps*

$$G_{\text{SM}} \longrightarrow G', \quad n_g \longrightarrow n'_g, \quad Y_{u,d,e} \longrightarrow Y',$$

stay inside $\mathcal{P}$ whenever G' is compact and the new spectrum is anomaly-free. None of G_{SM} , n_g , or the Yukawas appears in (??) or in the FGHMV source. The certified principles therefore do not select a unique element of $\mathcal{P}$. [P] (invariance).

Theorem ?? is the parameter-count test of this question. The first law is one structural constraint on the bulk metric. The matter ledger of the NSM is a finite list of installed choices. The space of holographic CFTs is infinite-dimensional (spectra, OPE coefficients, flavour groups). N_{free} exceeds $N_{\text{constraint}}$. Uniqueness fails.

4 What is selected

Given a holographic pair, the certified principles do select the bulk gravitational sector at linear order about AdS: the metric equation is Einstein [?], and the torsion coupling, if spinors are present, is the axial Einstein–Cartan one of Paper II (with the axial obstruction of Paper 14 standing at second order). That is “certify, select, assemble” applied to gravity, which is what Papers I–III already claimed.

Q4a asked for the same operation on the pair itself. The operation does not extend. Gravity is selected *given* a pair. The pair is not selected.

5 Existence remains open

Bottom-up, (??) in AdS *is* a holographic model. That is the same effective field theory, not a microscopic pair.

Top-down, a string or M-theory vacuum whose infrared is Palatini Einstein–Cartan plus the Standard Model is not constructed here and is not selected by entanglement first-principles. Citing an existing compactification as “the” pair would be a fabrication of uniqueness.

Paper 15’s quadratic axial deformation is a change of the bulk ultraviolet, not a choice of CFT.

Admission. I did not select a holographic pair. I proved that the program’s own principles cannot select one, and that the Standard Model cannot be one. Existence of some pair whose bulk infrared is the NSM is still [O]. That is the proper-context answer to Q4a.

6 Machine check

Solver `m9_5.q4a.pair.py`; auditor `m9_5.audit.q4a.py` (species table, no solver import). Gates C1–C5 PASS. The auditor recovers (b_1, b_2, b_3) by summing representations generation-by-generation. C3 is structural (the CHM integrand), not a numerical search.

7 Verdict

Claim	Status
Q4a as “invent a CFT”	refused (Paper 11)
Reading B (SM is the boundary CFT)	false, Thm. ??
Selection-uniqueness of the pair	false, Thm. ??
SM one-loop (b_1, b_2, b_3)	computed, audited
Bulk gravity given a pair	already certified, Papers I–III
Existence of some pair	still [O]
Final Status “UV completion”	unchanged, [O]

The phrase that is true: *Q4a, asked as a first-principles selection of the holographic pair, has a negative answer.* The phrase that is false: *we have the ultraviolet completion.*

References

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